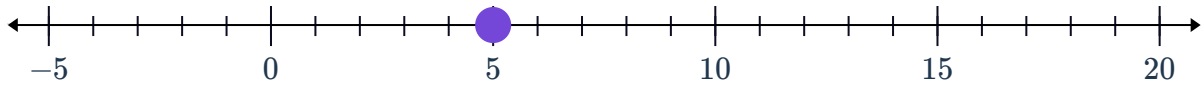


Addition of integers

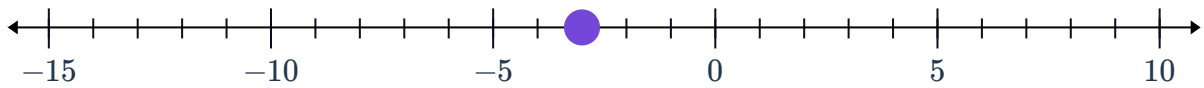


 1 Use the plotted point on the number line to add the following integers:

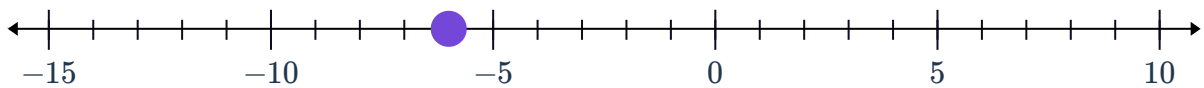
a $5 + 9$



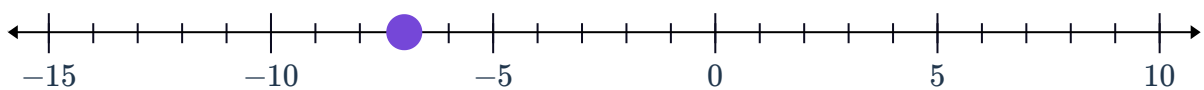
b $-3 + 8$



c $-6 + 3$



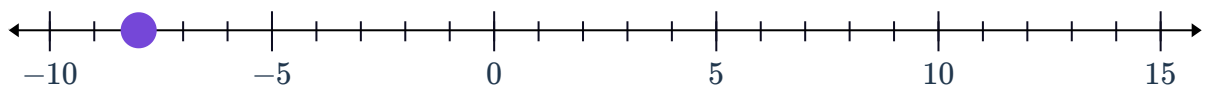
d $-7 + 7$



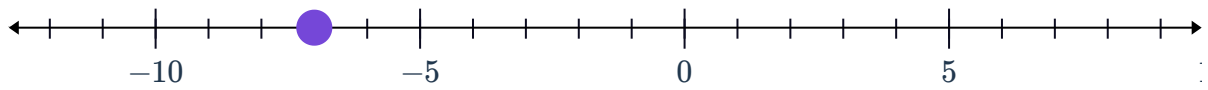
e $3 + 9 + 7$



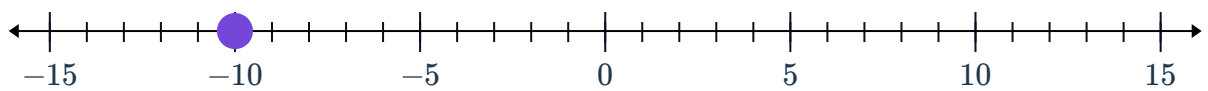
f $-8 + 6 + 9$



g $-7 + 7 + 4$



h $-10 + 5 + 12$



 2 Evaluate:



a $14 + 7$

b $-7 + 15$

c $-15 + 6$

d $-15 + 15$

e $-9 + 0$

f $-39 + 97$

g $-26 + 15$

h $-45 + 45$

i $-13 + 8 + 2$

j $-24 + 10 + 54$

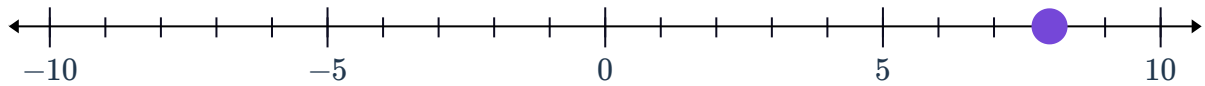
k $-21 + 0 + 21$

l $-88 + 28 + 92$

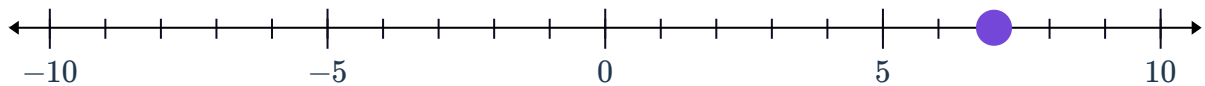
Subtraction of integers

 3 Use the plotted point on the number line to subtract the following integers:

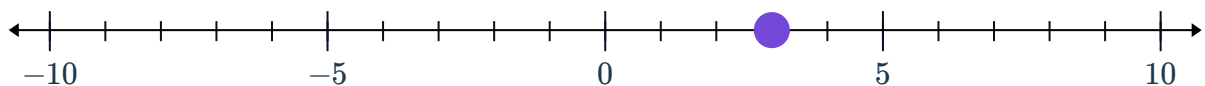
a $8 - 5$



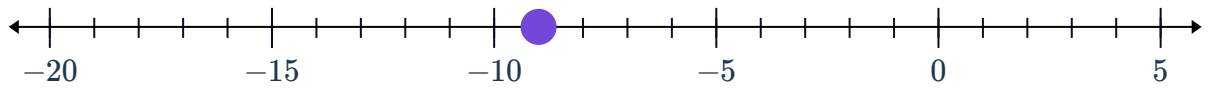
b $7 - 9$



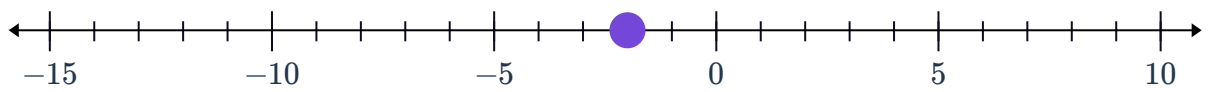
c $3 - 3$



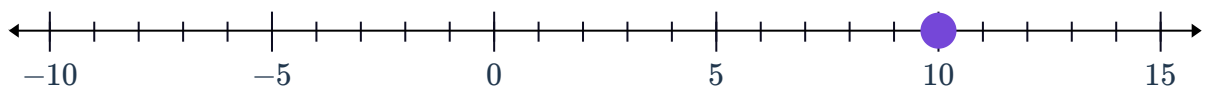
d $-9 - 9$



e $-2 + 7 - 4$



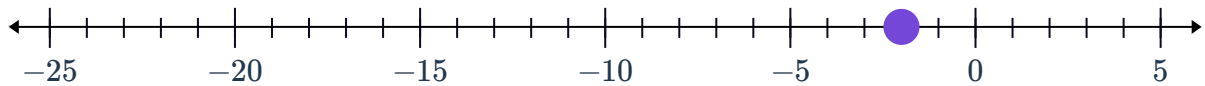
f $10 - 7 - 2$



g $14 - 18 + 1$



h $-1 - 15 - 3$



4 Evaluate:

a $12 - 3$

b $9 - 13$

c $11 - 11$

d $-13 - 13$

e $-32 - 41$

f $36 - 53$

g $-9 - 5 - 8$

h $-12 - 3 + 16$

i $27 - 19 - 10$

j $0 - 11 + 36$

k $-7 - 8 - 15$

l $14 - 49 + 35$

5 Evaluate:

a $7 + (-5)$

b $-7 + (-8)$

c $2 - (-3)$

d $-3 - (-9)$

e $-12 - (-12)$

f $17 + (-14)$

g $29 - (-52)$

h $-30 - (-32)$

i $12 + 17 + (-15)$

j $-12 + 16 + (-18)$

k $17 - 27 + (-17)$

l $-5 + (-2) + (-7)$

m $-9 - 2 - (-8)$

n $-5 - (-2) - (-10)$

o $-17 + 23 + (-20)$

p $34 - 63 - (-34)$

Applications

6 Valentina dives off a platform that is 8 m above the water. She descends 14 m before returning to the surface.

- If Valentina's initial position on the diving platform is represented by the directed number 8, find the directed number that represents her lowest position throughout the dive.
- Hence, state the greatest depth that Valentina reaches during the dive.

7 One evening in Moscow the temperature fell to -3°C . By midday the next day, the temperature was forecast to be 15°C . Find the expected rise in temperature.



- 8 Liquid nitrogen can be used to make ice cream. The ice cream mixture starts at 26°C , then rapidly cools to -185°C after liquid nitrogen is added to it. State the directed number that represents this change in temperature.
- 9 Three contestants are heading into the final round of a game show. Eileen has 39 points and Roxanne has -3 points. If Oprah has 14 more points than Eileen, calculate how many points separate Roxanne and Oprah.
- 10 A submarine is 148 m below sea level. It rises 34 m, then dives 47 m. Find the distance of the submarine below sea level.
- 11 Each bottle of orange juice is supposed to contain 250 mL. Nine bottles are inspected by the quality assurance department. The results of the inspection are shown in the table:

Bottle	1	2	3	4	5	6	7	8	9	10
Amount below or above the required volume (mL)	+4	-7	-11	+9	+3	0	-1	-5	+6	+1

- a Find the actual volume of:
- i Bottle 1 ii Bottle 2 iii Bottle 6 iv Bottle 10
- b Calculate the total volume contained in the 10 bottles.
- 12 Each letter has been assigned to an integer. A word must be formed using these letters and each letter can only be used once. The table shows the assignment of the integers:

Letter	A	E	I	C	T	N	R	S
Integer	-3	-2	-1	0	1	2	3	4

The score of the word is the sum of the integers assigned to its letters. Find the score of the following words:

- a NET b EAR c RAIN d CARES